Blind spot 08: learned branching for WDSat / XOR-CDCL on Semaev CNFs

Verdict: MATCHES (level: constant; no learned rule advances over the incumbent order)

Frozen rule (PROTOCOL.md), applied mechanically by summarize (card-10-r2, receipt raw/receipt-card-10-r2.json, status completed). Held-out cells n = 31, 33, 35, 37, l = 5, 128 targets (64 SAT planted, 64 UNSAT random), incumbent a = WDSat static order, cost D + P + Q + k W at k = 0, 0.1, 1 (the section 5a band applied to the native scorer work W), median of per-target ratio a / arm, 95% paired bootstrap (10,000 resamples). No arm reaches the 1.3 threshold at any weight, and none loses by the REGRESSES test (median <= 1/1.1 with interval below 1 at k = 0).

arm	k=0 median [95% CI]	k=0.1	k=1	rule verdict
b, learned static order (what -g can host)	0.9943 [0.9932, 0.9954]	0.9943 [0.9932, 0.9953]	0.9940 [0.9929, 0.9951]	MATCHES
b2, learned dynamic scorer	1.0013 [1.0000, 1.0013]	0.8980 [0.8584, 0.8985]	0.4669 [0.3856, 0.4684]	MATCHES
c, learned cube selection	1.0002 [1.0002, 1.0003]	0.9973 [0.9973, 0.9973]	0.9713 [0.9713, 0.9714]	MATCHES

Attribution is moot (nothing passed). The point estimates are within 0.6% of 1 at k = 0 for all three learned arms. The first run's validation hint (b_lin 1.36x fewer nodes on the 16 model-selection instances at n = 23..29) did not survive held-out targets: it was a selection effect on a small set (see "First run (superseded)").

Level: **constant** (a branching rule changes the cost of a tree walk, not the exponent). Gap to matched rho: **unchanged**. This is a stage diagnostic on toy fields: it computes no S, no rho ratio and no GAE total, so no cross-method row exists and the section 5a ranking does not apply; the 0 / 0.1 / 1 pricing above is applied to the arms' own native work and shows the dynamic scorer (b2) getting worse as its work is priced (0.90 at 0.1, 0.47 at 1). No distance to rho was reduced; no improvement over the index-calculus baseline is claimed.

Method

Native Rust replica of the WDSat search on the S'_4 formulation (instrument sha256 02478185...a280a, unchanged), validated against 60 committed stock WDSat decision counts (card 01: 60/60 decisions, 60/60 verdicts, 30/30 certificates). Learners (ridge, boosted trees) fitted on n = 23..29 (card 04), the committed raw/model.txt (sha256 00f0d29d...b60809) pinned in binds of cards 05-09b. Arms: a incumbent; a2 structural (bit-interleaved) order; b, b2, c learned; d random; bs, b2s shuffled-label controls; e CryptoMiniSat 5.11.21. Cards: 05-r2, 06-r2 (test), 07-r2, 08-r2 (secondary l = 6), 09-r2, 09b-r2 (CMS), 10-r2 (summaries). All six stage-1 receipts and card-10-r2 read completed; receipt sha256 values in the relays match the files on disk.

Numbers (primary, medians over 128 targets; unitless search events)

arm	median D	median P	median Q	median W
a	26704	3645315	0	26712
a2	30340	4036248	0	30340
b	28434	3887184	717	30048
b2	28503	3907093	0	4625662
c	28433	3876416	11751	152115
d	26744	3783728	0	26744
bs	30454	4115006	717	32057
b2s	27841	3891927	0	4586238

No target capped (0 of 1024 rows). By size the k = 0 ratio of b is 0.991, 0.995, 0.996, 0.994 for n = 31..37 (full per-size and per-population tables in raw/summary-primary-r2.txt). By population the median ratio of b is 0.995 on R (unsatisfiable) targets and 0.635 on P (planted) targets: on satisfiable targets the cost is dominated by where the first model lies, and the learned orders found it later (descriptive, from the summary; ratios of means over P targets computed from the raw rows, derived: b 0.83, b2 0.90, c 0.87 against a; the structural a2 0.63, random d 0.85). On R targets: b 0.995, b2 1.001, c 1.003, a2 1.022, d 0.959.

Controls: the structural order a2 matches a (1.0215 [1.0212, 1.0216]); random d is 4% worse (0.958 [0.934, 0.967]), so order matters by only a few percent on these trees (unsatisfiable trees have about 0.94 of 2^(3l) leaves for every order). Shuffled-label arms: bs 1.022, b2s 0.958 at k = 0, i.e. noise-fitted scorers do as well as the learned ones: the learned models carry no measurable signal. Positive controls (replica fidelity, planted targets all SAT, all arms agreeing with the exact count under --truth) held in every cell; the cards would have aborted.

Secondary (outside the rule): l = 6, n = 31 and 37, 3 P + 3 R each, 12 targets

b vs a 0.9963 [0.9955, 2.5414], b2 1.0004 [0.9675, 1.0004] (0.914 at k = 0.1, 0.505 at k = 1), c 1.0001 [1.0001, 1.9006]; all MATCHES. The intervals reach above 1.3 only through the three planted targets (12 pairs, wide). Not a result; consistent with the primary.

Arm (e), CryptoMiniSat 5.11.21 (descriptive, never in the rule; 3 P + 3 R at n = 31)

Built in-card at revision 4c9a6b6b...1794b. Median ratio a / e = 0.621 at k = 0 (CI [0.116, 3.720], 6 pairs; P 0.379, R 0.863). CMS needed a median 114,104 decisions against 25,700 for a (CMS prints rounded counters), and 16-23 s per UNSAT instance against about 0.4 s for the replica. Six targets cannot rank them; no claim.

Deviations and caveats

* Deviations 1-11 of PROTOCOL.md apply (rustc build, l fixed at 5, replica instead of the stock

  binary, repair pass with CMS built in-card). Deviation 11 records the repair.
* Card-10-r2 was executed directly with python3 tools/task_card.py run (no executor relay),
  after make_card_10_r2.py froze and linted it (CARD OK) on committed inputs.
* Relay files raw/relay-r2-*.txt are abbreviated copies (header and per-command status lines;
  rows elided), so check-relay reports relay_verbatim: false for all six stage-1 cards with
  no verify findings. This is a storage artifact of the assessor, not a finding of executor
  deviation: relay receipt hashes, commits and statuses equal the receipts on disk.
* Host: shared 4-vCPU container (L0, no isolation); wall times descriptive only. Toy scale
  (l = 5, n <= 37); the held-out axis is n and target, not l. A result for l ~ n/3
  fields is not implied. Learned models were fitted on a small sample (24 root states, 42 states
  per size).
* Not a speed result: no benchmarks/ entry, no leaderboard change.

Files

graph-primary-r2.svg/png (source make_graph_r2.py, from raw/summary-primary-r2.txt),
REPORT.pdf, raw/summary-{primary,secondary,cms}-r2.txt, raw/test-n*-r2.jsonl,
raw/sec-l6-n*-r2.jsonl, raw/cms-n31-*-r2.jsonl, raw/receipt-card-*-r2.json,
card-10-r2.yaml.

First run (superseded)

Superseded by the repair pass above (cards 05-09 were refused at lint because raw/model.txt
was not pinned; card 10 failed for lack of inputs). Kept for the record.

Verdict: BLOCKED (level: constant; primary test not run)

The frozen decision test (cards 05-10: held-out n = 31, 33, 35, 37, 128 targets,
arms b, b2, c against incumbent a, at 0 / 0.1 / 1 GAE per native operation)
never executed. Cards 05-09 were refused by task_card.py at lint: each
argv names raw/model.txt (written by card 04) but the card's binds do not
pin it, so the card cannot be frozen before the file exists. Card 10 ran its
build, then summarize failed because raw/test-n31.jsonl does not exist
(its inputs are produced by cards 05-08). No receipt exists for cards 05-09.
No ADVANCES/TRADES/MATCHES/REGRESSES claim is made. No gap-to-rho row exists
(no cross-method comparison was reached; §5a pricing at 0/0.1/1 GAE not applicable).

Restart of the builds was requested; the frozen cards cannot be repaired by
this step. Fix needed (director): re-freeze cards 04-10 with the card-04 output
model.txt (sha256 00f0d29d6f0535b91ca3ab0bbc1c76dbb19085fce325c5b77aabed0479b60809)
pinned in binds, and card 10 with its input files and build product handled
the same way, then re-dispatch 05-10.

What did run (measured)

card	content	outcome
01	selftest, rustc build, 60-instance corpus replay	selftest OK; replica == stock WDSat
decisions 60/60, verdicts 60/60, certificates 30/30
02	training samples n=23, 25	receipt status completed; samples-n23/25.txt on disk (executor
relay claimed non-completion; the receipt and files show it finished: relay deviated, reported not
repaired)
03	training samples n=27, 29	completed
04	fit ridge (lin) + GBT, validation	completed; raw/model.txt

Note: samples-n23/25/27/29.txt are all 78123 bytes (same instance/state
counts per size by design: 42 states per size, depths 0..8); they were not
checked to be distinct beyond differing content by size. Not independently audited.

Exploratory numbers (card 04 validation, NOT the frozen test)

16 validation instances at n = 23..29 (training-adjacent sizes, 2P+2R each),
mean log2 cost (search-tree nodes, native scorer work not priced):

- identity (incumbent order): 20.2961
- b_lin 19.8471 (-0.449 bits, about 1.36x fewer nodes); b_gbt 20.3625 (+0.07)
- b2_lin 20.5642, b2_gbt 20.6178 (worse)
- cube arms: 19.85 to 20.38 (best: full=gbt dcut=6 k=5 at 19.8457; worst lin dcut=6 k=3 at 20.3788)
- mean oracle gain of the best first variable over the incumbent's first
  variable at the root: 1.0036x (24 root states): the incumbent's first choice
  is already nearly optimal at the root.

No interval was computed, the validation set is the model-selection set (16
instances, one seed), the scorer's native work is unpriced (k = 0 only), and
no shuffled-label control or a2 comparison was run. These numbers are
descriptive and cannot support an advance. A gain of 1.36x by b_lin would
at best be a k = 0 constant-factor candidate, to be tested by the frozen rule.

Controls and deviations

Planned controls (bs, b2s shuffled labels, d, a2) were never run. Deviations
from the build notes are in PROTOCOL.md (rustc, ell = 5, 16P+16R targets,
native replica arm a). Additional deviation here: relay files for cards 1-3
are abbreviated copies of the executor text (data rows elided), so
check-relay reports relay_verbatim: false for them; receipts are
authoritative.
Caveats

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Host is L0-level (shared 4 vCPU container, no isolation); toy scale (ell = 5,
n <= 37); CMS 5.11.21 arm (card 09) not run; no wall-clock claims.

### Files

graph-validation.svg/png (recorded fit values; source make_graph.py +
raw/fit.txt), REPORT.pdf, raw/relay-1..10.txt, raw/receipt-card-*.json.
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